Math 534 H - Spring 2026 - Lecture 2

Some Examples of Nonlinear Equations

(8) Burgers' Equation (quasilinear - 1st order)

$$u_t + c\,u\,u_x = 0 \qquad x \in \mathbb{R}$$

inviscus version.

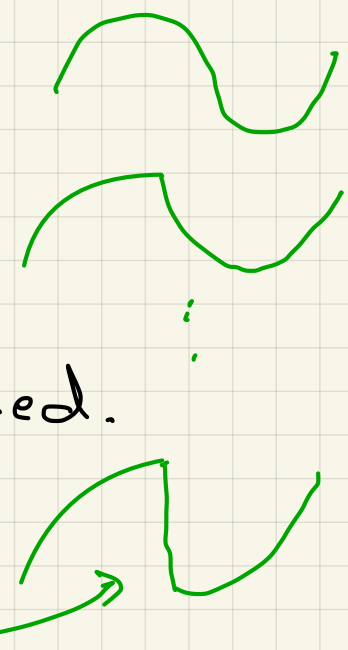
It governs a one dimensional flux of a non viscous fluid.

But it is also used to model traffic flow.

The viscous variant of this equation.

$$u_t + \underbrace{c\,u\,u_x}_{\text{Shock formation term}} = \underbrace{\varepsilon\,u_{xx}}_{\text{dissipation added.}}$$

Steepening term.



(9) <u>Porous medium equation</u> (2) (quasilinear; 2nd order)

$$u_t = k \cdot \operatorname{div}(u^\gamma \cdot \nabla u) \qquad k > 0 \,;\ \underline{\gamma > 1}$$

[<u>Note</u> if $\gamma = 0$ $\quad \operatorname{div}(\nabla u) = \Delta u$ e.g. $u_t = k \Delta u$
$u^0 = 1$ $\qquad$ linear heat equation]

This equation in the context of filtrations; such as modeling motion of water as it permeates in the ground.

(10) $$\underline{\underline{i}}\, u_t + \Delta_x u = \lambda |u|^2 u \qquad (\lambda \cdot u \bar{u} u)$$

(semilinear 2nd order)

$\lambda = \pm 1$ $\quad$ Cubic nonlinear

$$|u|^2 = u \cdot \bar{u}$$

$$|u(x,t)|^2 = u(x,t) \cdot \overline{u(x,t)}$$

Schrödinger equation (fiber optics; BEC; water waves, plasma physics, etc).

This is a <u>DISPERSIVE</u> <u>Equation</u>

$\lambda = 1$ DEFOCUSING, $\lambda = -1$ FOCUSING

(11) <u>Minimal surface equation</u> (quasilinear 2nd second order)

$$\operatorname{div}_x \left(\frac{\nabla_x u}{\sqrt{1+|\nabla u|^2}} \right) = 0$$

$x \in \Omega$ $\quad \partial\Omega$ is closed curve

$\Omega \subseteq \mathbb{R}^2$

(no time)

boundary

The graph of the solution u to this equation minimizes the area among all surfaces $z = V(x_1, x_2)$ whose boundary is a given (closed) curve.

<u>Ex.</u> soap bubbles are minimal surfaces.

(12) An example from geometric optics is the

<u>Eikonal Equation</u> $\quad |\nabla_x u| = c(x) \quad x \in \mathbb{R}^d$.

Fully nonlinear; 1st order $\quad x \in \underline{\underline{\Omega}} \subseteq \mathbb{R}^d$

TRANSPORT EQUATIONS WITH CONSTANT COEFFICIENTS

on $(t, x) \in \mathbb{R}^2$

$$u_t + \underline{\underline{c}}\, u_x = 0$$

(c constant)

Q: How do we solve this?

We will be able to prove that solutions exist by direct "construction" (explicitly).

Before we do that let's understand where this equation comes from?

Consider a fluid (water) flowing at a constant rate $\underline{\underline{c}}$ along a horizontal pipe of fixed

$$a u_x + b u_t = 0$$

$a, b \neq 0$

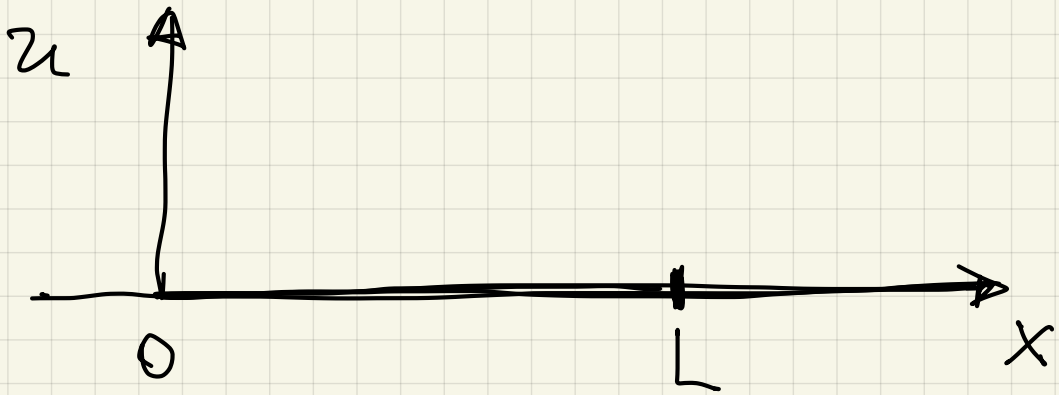
$$u_t + \underbrace{\frac{a}{b}}_{= c} u_x = 0$$

$t \leftrightarrow y \quad (x, y) \in \mathbb{R}^2$

$$a u_x + b u_y = 0$$

All the same.

Cross section in the positive x-direction



A substance (pollutant/dye) is suspended in the fluid; and

$u(x,t)$ = concentration (gm/cm) at time t and position x. <u>If diffusion is negligible</u> then what happens is that the rate of change u_t of concentration is proportional to u_x (or more generally in higher dimensions to the $\nabla_x u$). This says (†)

(†) $$\boxed{u_t + c\,u_x = 0}$$

$u_t = u_t(x,t)$

$u_x = u_x(x,t)$.

We'll explain (†) later but assuming this.

How do we solve it?

$$\underbrace{}\, u_t + \underbrace{c}\, u_x = 0 = \underbrace{c u_x} + \underbrace{u_t}$$

$$\vec{v} = (c, 1) \qquad = \nabla_{xt} u \cdot \vec{v}$$

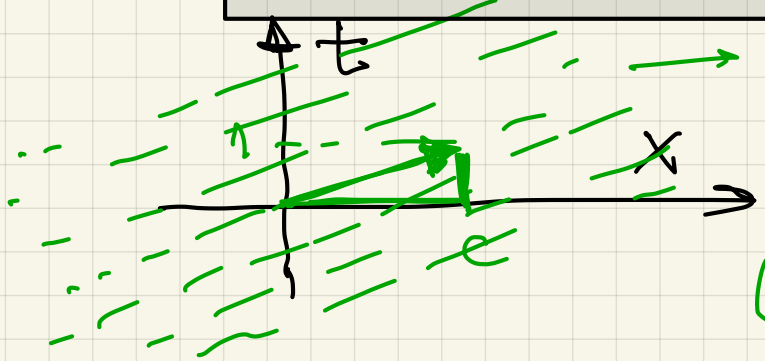
$$\nabla_{xt} u = (u_x, u_t) \qquad = D_{\vec{v}}\, u$$

directional derivative of $u(x,t)$ in the direction of $\vec{v}$

What we get is that

$$D_{\vec{v}}\, u = 0 \iff \boxed{u \text{ is constant in the direction of } \vec{v}}$$

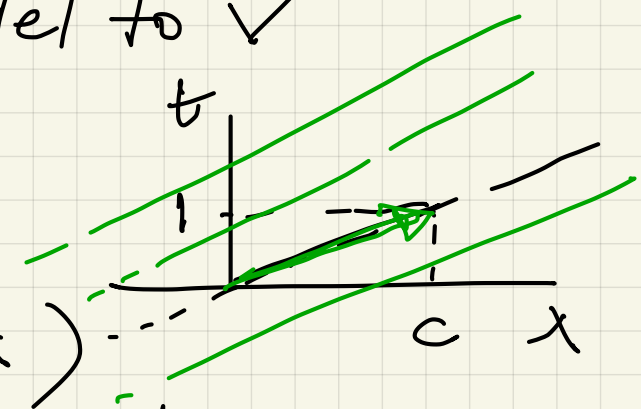
Slope of $\vec{v} = \frac{1}{c}$



lines parallel to $\vec{v}$. All this lines have the same (slope as $\vec{v}$) $= \left(\frac{1}{c}\right)$

The equation of the lines parallel to $\vec{v}$ (called <u>characteristic lines</u>)

(constant coefficients)



are

$$x - ct = \underline{\underline{\text{constant}}} \quad \text{(all parallel)}$$

$$t = \frac{1}{c}x$$

$$x - ct = 0$$

(the lines move as the constant changes)

Then if the solution $u(x,t)$ is constant in the direction of $\vec{v}$ we must have that u only depends on $x - ct$ $\Longleftrightarrow$

$$\boxed{u(x,t) = f(x-ct)}$$ for any

f function of 1 variable.